Structure:

In phosphine, phosphorus shows sp^3 hybridisation. Three orbitals are occupied by bond pair and fourth corner is occupied by lone pair of electrons. Hence, bond angle is reduced to 93.5°. Phosphine has a pyramidal shape.

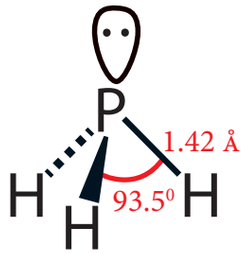


Figure 3.4 Structure of phosphine

Uses of phosphine:

Phosphine is used for producing smoke screen as it gives large smoke. In a ship, a pierced container with a mixture of calcium carbide and calcium phosphide, liberates phosphine and acetylene when thrown into sea. The liberated phosphine catches fire and ignites acetylene. These burning gases serves as a signal to the approaching ships. This is known as **Holmes signal**.

3.1.10 Phosphorous trichloride and pentachloride:

Phosphorous trichloride:

Preparation:

When a slow stream of chlorine is passed over white phosphorus, phosphorous trichloride is formed. It can also be obtained by treating white phosphorus with thionyl chloride.

$$P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$$

Properties

When phosphorous trichloride is hydrolysed with cold water it gives phosphorous acid.

$$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$$

This reaction involves the coordination of a water molecule using a vacant 3d orbital on the phosphorous atom following by elimination of HCl which is similar to hydrolysis of $SiCl_4$.

$$PCl_3 + H_2O \longrightarrow PCl_3.H_2O \longrightarrow P(OH)Cl_2 + HCl$$

This reaction is followed by two more steps to give $P(OH)_3$ or H_3PO_3.

$$HPOC1_2 + H_2O \longrightarrow H_2PO_2Cl + HCl$$
$$H_2PO_2Cl + H_2O \longrightarrow H_3PO_3 + HCl$$

Similar reactions occurs with other molecules that contains alcohols and carboxylic acids.

$$3C_2H_5OH + PCl_3 \longrightarrow 3C_2H_5Cl + H_3PO_3$$
$$3C_2H_5COOH + PCl_3 \longrightarrow 3C_2H_5COCl + H_3PO_3$$

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